

## Lesson plan

|                                          |                                                |                                    |
|------------------------------------------|------------------------------------------------|------------------------------------|
| <b>Name of the Programme:</b>            | <b>BCA/BSc</b>                                 |                                    |
| <b>Title of the Course:</b>              | <b>Elements of Linear Algebra</b>              |                                    |
| <b>Course code</b>                       | <b>CS1831</b>                                  |                                    |
| <b>Semester:</b>                         | <b>I</b>                                       |                                    |
| <b>Names of the Course Instructor(s)</b> | <b>Dr. Ankita Chandola<br/>Dr. Akhilesh KS</b> |                                    |
| <b>Course credit</b>                     | <b>No. of hours per week<br/>(1+1+0)</b>       | <b>Total no. of Teaching hours</b> |
| <b>02</b>                                | <b>02 Hours</b>                                | <b>30 Hours</b>                    |

| <b>Sessio<br/>n</b> | <b>Modul<br/>e No.</b> | <b>Topic</b>                                                           | <b>Pedagogy /activities</b> | <b>Pre-<br/>reading/referen<br/>ce</b>                             |
|---------------------|------------------------|------------------------------------------------------------------------|-----------------------------|--------------------------------------------------------------------|
| 1.                  | <b>1</b>               | Introduction to vectors, types of vectors                              | Interactive Lecture         | Chapter 1.1,1.2, Gilbert Strang                                    |
| 2.                  |                        | Practice Questions on Vectors                                          | Interactive Lecture         |                                                                    |
| 3.                  |                        | Vector Operations: Dot and Cross Product, Coplanarity.                 | Board work + Examples       |                                                                    |
| 4.                  |                        | Practice Questions on Dot product, and cross product, coplanarity      | Interactive Lecture         |                                                                    |
| 5.                  |                        | Linear Combinations, Dependence & Independence                         | Board work + Examples       |                                                                    |
| 6.                  |                        | Practice Questions on Linear combinations, dependence and independence | Interactive Lecture         |                                                                    |
| 7.                  | <b>2</b>               | Matrix types, Matrix Operations                                        | Board work + Examples       | Gilbert Strang Chapter 1.3, chapter 2.<br><br>Chapter 2, BS Grewal |
| 8.                  |                        | Practice Questions on Matrix Operations                                | Interactive Lecture         |                                                                    |
| 9.                  |                        | Determinants, Minors and Cofactors                                     | Board work + Examples       |                                                                    |
| 10.                 |                        | Practice Questions on Determinants, Minors and cofactors.              | Interactive Lecture         |                                                                    |
| 11.                 |                        | Adjoint, Inverse of Matrices                                           | Board work + Examples       |                                                                    |
| 12.                 |                        | Practice Questions on adjoint, inverse of matrices.                    | Interactive Lecture         |                                                                    |

|     |   |                                                                     |                       |                                      |
|-----|---|---------------------------------------------------------------------|-----------------------|--------------------------------------|
| 13. | 3 | Rank of a Matrix, Echelon Form                                      | Board work + Examples | Chapter 2, Gilbert Strang            |
| 14. |   | Row operations, Gaussian elimination                                | Board work + Examples |                                      |
| 15. |   | Solving Systems: Gauss, Inverse, Row Canonical                      | Board work + Examples |                                      |
| 16. |   | Solve 3x3 systems, row reduction drills                             | Interactive Lecture   |                                      |
| 17. |   | Gauss Jordan Method                                                 | Board work + Examples |                                      |
| 18. |   | Practice Questions on row reduction                                 | Interactive Lecture   |                                      |
| 19. | 4 | Linear Transformations and their Properties                         | Board work + Examples | Chapter 3, chapter 8, Gilbert Strang |
| 20. |   | Definition, matrix form of transformations                          | Board work + Examples |                                      |
| 21. |   | Basis and Dimension                                                 | Board work + Examples |                                      |
| 22. |   | null space, column space                                            | Board work + Examples |                                      |
| 23. |   | Rank Nullity Theorem                                                | Board work + Examples |                                      |
| 24. |   | Practice Questions                                                  | Interactive Lecture   |                                      |
| 25. | 5 | Eigenvalues, Eigenvectors                                           | Board work + Examples | Chapter 6, Gilbert Strang            |
| 26. |   | Characteristic equation, Problems on eigenvalues and eigen vectors. | Board work + Examples |                                      |
| 27. |   | Diagonalization of Matrices                                         | Board work + Examples |                                      |
| 28. |   | Problem-solving on diagonalization                                  | Interactive Lecture   |                                      |
| 29. |   | Applications of Eigenvalues and Eigenvectors                        | Interactive Lecture   |                                      |
| 30. |   | Practice Questions                                                  | Interactive Lecture   |                                      |

### Assessment and Evaluation

| Continuous Internal Evaluation (CIE) :70% |                             |
|-------------------------------------------|-----------------------------|
| Components of CIE                         | Weightage of each component |
| CIE-1<br>- Quiz Assessment                | 20 marks out of 70          |

|                                          |                    |
|------------------------------------------|--------------------|
|                                          |                    |
| CIE-2<br>- Theory (Pen and Paper Theory) | 25 marks out of 70 |
| CIE-3<br>- Activity/Case Study/Tutorial  | 25 marks out of 70 |

|                                      |                      |
|--------------------------------------|----------------------|
| <b>Semester End Exams (SEE): 30%</b> |                      |
| Mode of Exam                         | Pen and Paper Theory |
| Weightage of exam                    | 30%                  |

### Course Outcomes- Programme Outcomes Matrix

| Program |                                                                                                | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 |
|---------|------------------------------------------------------------------------------------------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|
| Outcome | Course Outcome                                                                                 |     |     |     |     |     |     |     |     |     |      |      |      |
| CO1     | CO1: Apply vector algebra concepts to solve mathematical and engineering problems effectively. | 3   | 2   |     |     | 2   |     |     | 2   |     | 2    |      |      |
| CO2     | CO2: Apply matrix algebra and determinants to solve complex mathematical problems.             | 3   | 2   |     |     | 2   |     |     | 2   |     | 2    |      |      |
| CO3     | CO3: Solve systems of linear equations using various methods.                                  | 3   | 2   |     | 2   | 2   |     |     | 2   |     | 2    |      |      |

|     |                                                                                                     |   |   |  |   |   |  |  |   |  |   |  |
|-----|-----------------------------------------------------------------------------------------------------|---|---|--|---|---|--|--|---|--|---|--|
| CO4 | CO4: Solve problems involving calculation related to eigenvalue and eigenvector for a given matrix. | 3 | 2 |  | 2 | 2 |  |  | 2 |  | 2 |  |
|-----|-----------------------------------------------------------------------------------------------------|---|---|--|---|---|--|--|---|--|---|--|

### Rubrics for Assessment Components

| Assessment | Component       | Type of Component                        | Rubrics for Assessment                                                                                                                                  |
|------------|-----------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1          | CIE1 (20 Marks) | Quiz Assessment (20 Marks)               | A quiz will be conducted, consisting of problem-solving. Evaluation is done according to the Scheme of Evaluation prepared by a team of course faculty. |
| 3          | CIE2 (25 Marks) | Mid Sem Exam                             | The test will be conducted out of 25 marks and will be evaluated according to the Scheme of Evaluation prepared by a team of course faculty             |
| 4          | CIE3 (25 Marks) | Activity/Case Study/Tutorials (25 Marks) | Can be group or individual work; Questions/real-world applications using pen and paper or presentation.                                                 |
| 5          | SEE             | SEE (30 Marks)                           | The test will be conducted out of 30 marks and will be evaluated according to the Scheme of Evaluation prepared by a team of course faculty.            |

### Rubrics for each component of CIE

| CIE Components | Excellent (5 Marks) | Very Good(4) | Good(3) | Average (2 Mark) | Needs Improvement (1 Marks) |
|----------------|---------------------|--------------|---------|------------------|-----------------------------|
|----------------|---------------------|--------------|---------|------------------|-----------------------------|

| Quiz/Tutorial<br>/Activity | Able to<br>solve/present<br>all problems<br>correctly                         | Able to<br>solve/present<br>all<br>problems<br>with<br>hints. | Able to<br>solve/present<br>most problem<br>but require<br>hints in some       | Able to<br>solve/present<br>all<br>problems only<br>after giving<br>hints | Unable to<br>solve/present<br>most problems<br>even<br>after hints. |
|----------------------------|-------------------------------------------------------------------------------|---------------------------------------------------------------|--------------------------------------------------------------------------------|---------------------------------------------------------------------------|---------------------------------------------------------------------|
| Mid<br>term/SEE)           | Able to solve<br>all<br>L3 and L2<br>level<br>problems<br>without<br>mistakes | Able to<br>solve all<br>L2 and L3<br>problems<br>partially.   | Able to solve<br>L2 and L3<br>level<br>problems<br>partially with<br>mistakes. | Able to solve<br>less<br>than 50% of<br>the<br>problems with<br>mistakes  | Able to solve<br>only<br>partially and<br>with<br>mistakes          |

**Dr. Ankita Chandola**



**Signature of the Course Faculty**